

DATA 2010 Project Project Report: Trends in Olympic Athlete Performance and Representation (1896–2024)

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1 Project Overview

Research question (what we are trying to find out). The Olympic Games provide a long-running record of athlete participation and medal outcomes, allowing us to study how performance and representation evolve over time. Using the *Olympic Summer and Winter Games (1896–2024)* dataset (athlete–event records with variables such as **year**, **season**, **city**, **sport**, **event**, **team**, **sex**, **age**, and **medal**), this project investigates three connected questions. First, we examine how medal outcomes vary by **age group** and **sex** across sports, and identify which sports show the largest changes in **gender parity** and **peak-age patterns** across decades. Second, we evaluate how **country rankings** change when we switch from **athlete-medal counts** (row-based counts that can inflate team events) to **event-medal counts** (deduplicated medals per event), and we measure country **efficiency** as medals per athlete-event entry over time. Third, we identify which **sports and events** generate the most medal-winning athletes and analyze whether countries are becoming more **specialized** or **diversified** in their medal portfolios, including how stable their performance is across Olympic editions and whether host editions correspond to measurable shifts in medal distribution by sport.

Why it matters (who would care and why). The answers matter to stakeholders who make decisions about athlete development and sport investment. National sport organizations and coaches can use age- and sex-specific medal patterns to support talent pipeline planning (e.g., which sports peak earlier vs. later, and where parity is improving or lagging). Ana-

lysts, journalists, and policymakers can benefit from fairer country comparisons by understanding how medal totals depend on the counting method (athlete-medal vs. event-medal), especially for team sports. Finally, understanding which sports and events produce the most medal-winning athletes and whether countries trend toward specialization or diversification can inform long-term strategy, resource allocation, and expectations about stability versus volatility in Olympic performance.

Main result . [Write after completing analysis: briefly summarize 2–4 key findings that directly answer the three sub-questions, e.g., which sports showed the largest parity changes, which countries changed rank the most under event-medal counting and which were most efficient, and which sports/events dominated medal production along with evidence of specialization/diversification trends.]

2 Research Question (Make it Specific)

The Olympic Games are not only a global competition but also a long-running dataset that can reveal how representation, performance, and strategic investment in sport evolve over time. Using Olympic records spanning 1896–2024, this project investigates patterns with real-world relevance to athlete development and national sport planning. Specifically, we focus on three research questions:

Our 3 Sub-questions

- 1) How do medal outcomes vary by age group and sex across sports, and which sports show the largest changes in gender parity and peak-age patterns across

decades?

- 2) How do country rankings change over time when switching from athlete-medal counts to event-medal counts, and which countries are most efficient (medals per athlete-event entry) under each method across Olympic years?
- 3) Which sports and events generate the most medal-winning athletes, and how do countries' medal portfolios (specialization vs diversification) and performance stability change across Olympic editions, especially around host years where medal distribution by sport may shift?

3 Data Source and Description

3.1 Where the data came from

- **Dataset name:** Olympic Summer and Winter Games (1896–2024)
- **Link (Kaggle):** <https://www.kaggle.com/datasets/magicchris/olympic-summer-and-winter-games-1896-2024>
- **Dataset owner (as shown on Kaggle):** Cassini-Chris (Owner)
- **Authors:** Not specified on the Kaggle dataset page
- **Provenance:** Not specified on the Kaggle dataset page (no source/citation details provided).
- **Citations / DOI:** Not specified on the Kaggle dataset page.
- **What each row represents:** One row represents one athlete's record in one Olympic event for a specific Olympic Games, including the athlete's sex/age (if present), country/team, the sport and event, the Games year/season/city, and the medal outcome (if any).
- **Key columns used (2–8 columns):**
 - **Year** (Olympic year)
 - **Season** (Summer/Winter)
 - **Team** (country/team represented)
 - **Sport** (sport category)

- **Event** (specific event)
- **Sex** (athlete gender)
- **Age** (athlete age, if included in the file)
- **Medal** (Gold/Silver/Bronze/No medal)

- **License:** Apache License 2.0 (as listed on the Kaggle dataset page).
- **Expected update frequency:** Not specified on the Kaggle dataset page.

3.2 Basic dataset facts

Include a small table like this (fill with your values):

Number of rows	290,697
Number of columns	8
Time range (Year)	1896–2024
Seasons included	Summer & Winter
Unique years (Games editions)	38
Unique countries/teams (team)	1151
Unique sports (sport)	92
Unique events (event)	1534
Medalist rows (Gold/Silver/Bronze)	247,941
Age range (years)	10–72
Sex proportion (M/F)	69.38% / 30.62%
Missing values present?	No
Exact duplicate rows removed	[#]

4 Data Cleaning and Feature Engineering

This section documents how we prepared the Olympics dataset for analysis and created new variables to support the three research questions.

Minimum expectations (what applies here):

- Fixed data types (e.g., **year** and **age** converted to numeric).
- Handled missing values (imputed missing **age** using group medians).
- Removed duplicates / impossible values (removed exact duplicate rows; dropped rows with invalid **year**).

- Dealt with outliers *carefully* (flagged suspicious ages instead of deleting them).

What we actually did:

- Standardized column names (lowercase with underscores) and stripped whitespace from text fields (e.g., `team`, `sport`, `event`, `medal`).
- Converted `year` to numeric and removed rows with missing/invalid `year`, since time is required for all analyses.
- Converted `age` to numeric (invalid values coerced to missing) and examined the distribution for unusual values.
- Standardized `medal` values by filling missing medals as `No medal` and ensuring consistent labels for `Gold`, `Silver`, `Bronze`, and `No medal`.
- Removed exact duplicate rows to prevent double counting.
- Flagged suspicious ages using an indicator `age_flag_suspicious` (e.g., very young or unusually old ages) to avoid deleting outliers without justification.
- Imputed missing ages by filling `age` with the median age within each (`sex`, `sport`) group, with an overall median used as a fallback, and stored the result as `age_imputed`.
- Engineered new features:
 - `decade` = $\lfloor \text{year}/10 \rfloor \times 10$ to analyze long-term trends.
 - `age_group` (Olympic-style bins such as `<16`, `16--18`, `19--25`, ..., `51+`).
 - `won_medal` (binary indicator for Gold/Silver/Bronze).
 - `medal_points` (Gold=3, Silver=2, Bronze=1, No medal=0) for summary comparisons.
- Created an *event-medal* view for country-level medal counting by deduplicating medals at (`year`, `event`, `team`, `medal`) to reduce team-event inflation.

Show evidence (before vs. after):

Check	Before cleaning
<code>year</code> type	mixed / not guaranteed numeric
Missing <code>year</code> rows	present
<code>age</code> type	mixed / non-numeric values possible
Missing <code>age</code>	present
Exact duplicates	possible
Medal labels	inconsistent / missing possible
Outliers	unclear

5 Exploratory Data Analysis and Visualization

You should include **at least 4** well-labeled figures. Every figure must answer: *What should the reader learn from this? Tell a story about your data and the plot.*

5.1 Plot 1: [Title]

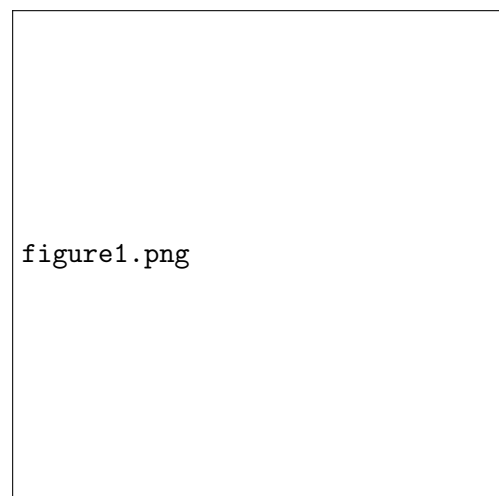


Figure 1: [What this figure shows. Mention the key takeaways.]

5.2 Plot 2: [Title]

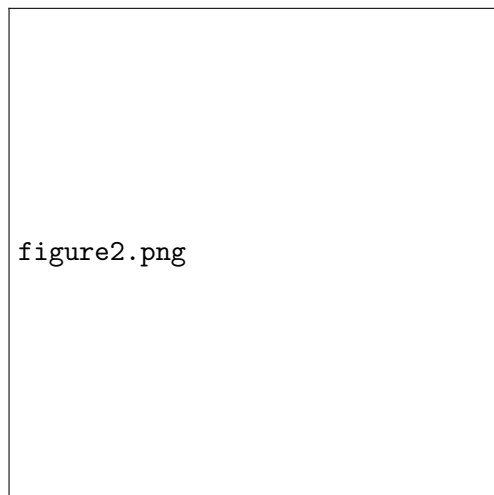


Figure 2: [Key takeaway.]

5.3 Plot 3: [Title]

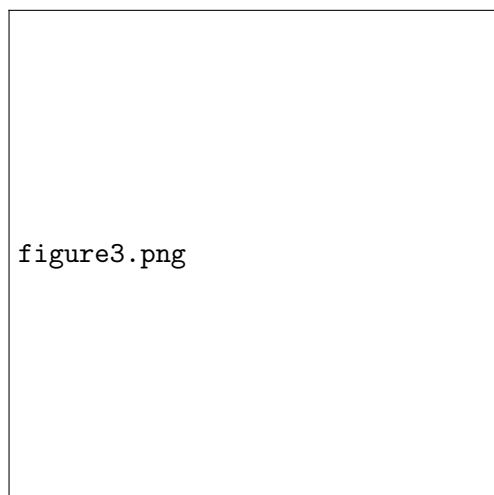


Figure 3: [Key takeaway.]

Common plot mistakes to avoid:

- Bar chart for a continuous variable (use histogram/density instead).
- No axis labels/units.
- Too many categories (group small ones into “Other” or filter).

6 Data Analysis (Techniques From This Course)

Pick a direction that fits your question. Use **multiple** techniques (not just one).

6.1 Analysis Plan

In 3–6 sentences: what methods did you use and why do they answer your question?

6.2 Baseline (If you do prediction)

If you predict something, include a simple baseline first (e.g., mean predictor, simple linear regression, majority class).

6.3 Main Results

Present results with at least **one table** and explain it in words.

Model / Method	Metric 1 (e.g., Accuracy)	Metric 2 (e.g., F1)
Baseline	[]	[]
Your method	[]	[]

If you used a train/test split: say how you split and why (and don’t evaluate on training data only).

6.4 Interpretation (So What?)

Don’t just report numbers. Explain:

- What the results mean in plain language,
- What surprised you (if anything),
- What limitations might change your conclusion.

7 Real-World Application

Your project should produce **useful insight** or a **decision** someone could act on.

Answer:

- Who is the stakeholder? (customer, city planner, student, business, etc.)
- What decision could they make based on your results?
- What would you recommend (1–3 bullets), and how confident are you?

8 Report Quality and Reproducibility

8.1 How to run your code

- Where your code is (GitHub/ZIP):
- Main file/notebook name:
- Steps to run (3–6 bullets):
- Required packages (if any):

8.2 Project structure (recommended)

- README.md (how to run)
- notebooks/ (analysis notebooks)
- src/ (helper functions, optional)
- figures/ (exported plots, optional)

9 Contributions (If Group Project)

List what each person did. Be honest and specific.

- Member 1:
- Member 2:
- Member 3:

10 Conclusion and Next Steps

Conclusion: 1 paragraph summarizing the answer to your question.

Next steps: If you had 1 more week, what would you improve or explore?

11 AI Disclosure

If you used AI tools (e.g., ChatGPT, Gemini) for coding or writing, disclose it.

- Tools used (if any):
- What you used them for (debugging, rewriting, brainstorming, etc.):
- Prompts you used:

References (No research papers required)

You do **not** need research papers. You should cite:

- Your dataset link (required),
- Any important documentation/tutorial you relied on,
- Any external code sources (if any).
- Dataset: [Name], [Link], accessed [date].
- Documentation: [Library/Tool], [Link].

A Appendix (Optional)

Extra figures, extra tables, or small code snippets that would clutter the main report.